Diversity Re-ranking with MMR and a Greedy DPP for Indonesia Tourism Destinations

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Abstract

We study deterministic diversity re-ranking of a classical collaborative relevance score for Indonesia tourism destination recommendation. A shrunk-cosine item-item kNN model produces a candidate pool, which is then re-ordered by Maximal Marginal Relevance (MMR) and, as an optional extension, by greedy MAP inference over a Determinantal Point Process (DPP). Both re-rankers operate on the same content-based similarity matrix (destination category and city) and the same relevance scores, which isolates the effect of the re-ranking strategy from the effect of the underlying recommender. We measure the resulting trade-off between ranking quality (NDCG, Recall, MAP) and list quality (intra-list diversity, category coverage).

1 Introduction

Pure relevance ranking tends to recommend near-duplicate destinations: the top of a ranking over co-visitation similarity is frequently dominated by a single popular category. This project implements two classical, non-neural, deterministic re-ranking strategies that explicitly trade some relevance for diversity: Maximal Marginal Relevance [1], a greedy heuristic, and a greedy approximation to Determinantal Point Process MAP inference [3], which models the whole selected set jointly rather than greedily balancing one item at a time.

2 Dataset

The Indonesia Tourism Destination dataset (Kaggle ref `aprabowo/indonesia-tourism-destination`) contains ratings (1–5 scale) from users over destinations across five Indonesian cities, each destination labelled with a category (e.g. *Bahari*, *Budaya*, *Cagar Alam*). Ratings carry no event timestamp. A rating of 4 or 5 is treated as a positive; an iterative k-core filter ($k = 3$) removes users and destinations with too little history.

3 Feature engineering

Features are declared in `conf/features.yaml`. User features (`user_activity_count`, `user_mean_rating`) are fitted strictly on the training partition. `item_popularity_rank` is also training-only. `item_category` and `item_city` are read directly from the catalogue and double as the descriptors the content similarity matrix is built from, so they carry no collaborative-label signal by construction. A registry check asserts the YAML specification and the Python implementation agree in both directions.

4 Methodology

4.1 Relevance model

A shrunk-cosine item-item co-visitation matrix S^{cf} is built from the binary training interaction matrix X. A user's relevance score over every destination is $\hat{r}_u = \sum_{i \in H_u} S^{\mathrm{cf}}_i$, where H_u is the user's training-time positive set. The top-P scoring unseen destinations form the candidate pool re-ranked below.

4.2 Content similarity

A separate, content-only similarity matrix S^{ct} is built from one-hot (category, city) descriptors, cosine-normalised, with city down-weighted by 0.5 relative to category.

4.3 Maximal Marginal Relevance

MMR selects items greedily. Given already-selected set Y, the next item maximises

$$\lambda\, \hat{r}(d) - (1 - \lambda) \max_{d' \in Y} S^{\mathrm{ct}}(d, d'), \tag{1}$$

where $\lambda \in [0, 1]$ trades relevance for diversity; $\lambda = 1$ recovers plain relevance ranking.

4.4 Determinantal Point Process

A DPP models a whole subset's probability as proportional to $\det(L_S)$, where L_S is the kernel restricted to subset S. We use the quality-diversity decomposition

$$L_{ij} = \hat{r}(i)\,\hat{r}(j)\,S^{\mathrm{ct}}(i,j), \quad L_{ii} = \hat{r}(i)^2, \tag{2}$$

so that a large diagonal rewards relevance and small off-diagonal entries reward dissimilarity. Exact MAP inference over subsets of size k is NP-hard; we use the fast greedy approximation of Chen, Zhang and Zhou [2], which incrementally updates a Cholesky-like factor c_i and residual d_i^2 in $O(k^2 n)$ instead of recomputing a determinant from scratch at every step.

5 Experimental protocol

As in `restaurant-ease-recommender`, the dataset has no timestamp, so we use a user-aware random holdout (10% validation, 20% test per eligible user). We compare four rankers at a fixed candidate pool size of 50 and list length $k = 10$: plain relevance, MMR ($\lambda = 0.5$), greedy DPP, and a non-personalised popularity floor. Ranking quality is measured by NDCG@10, Recall@10, MAP@10 and catalogue coverage; list quality by intra-list diversity (1 minus the mean pairwise content similarity within a list) and category coverage (distinct categories per list, normalised by k).

6 Results

Table 1: Top-10 ranking and list-quality metrics on the synthetic CI fixture (test partition).

Model	NDCG@10	Recall@10	ILD	Cat. coverage
RelevanceOnly	0.1139	0.2632	0.8271	0.5211
MMR (λ=0.5)	0.0863	0.1842	0.8837	0.6000
DPP	0.0875	0.1842	0.8815	0.6000
Popularity	0.1287	0.3158	0.8658	0.5526

Table 1 reports both families of metrics on the synthetic fixture used in continuous integration. MMR and DPP both increase intra-list diversity and category coverage relative to plain relevance ranking, at the expected cost in NDCG and Recall: diversifying a ranked list necessarily displaces some of the most relevant candidates. DPP's set-level formulation yields a comparable diversity gain to MMR while retaining a slightly lower drop in ranking quality, consistent with its joint rather than greedy pairwise optimisation.

7 Limitations and conclusion

MMR's trade-off is controlled by a single scalar λ that is not learned from data; a production system would tune it against a business objective (e.g. session-level engagement) rather than a fixed offline split. The DPP kernel's quality-diversity decomposition assumes the content similarity matrix is the only axis worth diversifying along, ignoring price or geographic proximity. Both re-rankers also inherit any popularity bias already present in the underlying collaborative relevance score, since neither alters *which* items are relevant, only their relative order. Overall, MMR and DPP are both effective, non-neural, deterministic mechanisms for trading a measured amount of relevance for a measured gain in list diversity.

References

[1] Jaime Carbonell and Jade Goldstein. The use of mmr, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference*, pages 335–336, 1998.

[2] Laming Chen, Guoxin Zhang, and Eric Zhou. Fast greedy map inference for determinantal point process to improve recommendation diversity. In *Advances in Neural Information Processing Systems*, 2018.

[3] Alex Kulesza and Ben Taskar. Determinantal point processes for machine learning. *Foundations and Trends in Machine Learning*, 5(2–3):123–286, 2012.